Ashwin Chembu

(341) 356-1722 | ashchembu@gmail.com | San Francisco, California

PROFILE

Applied-AI engineer building retrieval-augmented products with document ingestion, vector search, evidence extraction, and cited answers. Connects grounded AI workflows to usable interfaces and backend services.

TECHNICAL SKILLS

Languages / Product: Python, TypeScript, SQL, React, FastAPI, Flask, Node.js, MongoDB, Redis, AWS S3
AI / Retrieval: RAG, LangChain, LangSmith, OpenAI, Azure OpenAI, Pinecone, FAISS, NumPy
Inference / Quality: PyTorch, Hugging Face Transformers, document ingestion, vector search, pytest

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — **South San Francisco, CA | Jun 2024-Present**

- Built React/**FastAPI RAG** assistants with AWS document ingestion and Redis-backed services, connecting uploaded knowledge to searchable enterprise workflows so teams could retrieve context through usable product interfaces.
- Contributed **FastAPI**/Uvicorn model workflows and pytest checks for enterprise **RAG** services, connecting document-ingestion and evaluation tooling so backend behavior could be inspected alongside the product integration.
- Built Python/Flask endpoints connecting AWS S3 and Smartsheet APIs, coordinating document storage and workflow handoffs so teams could ingest, organize, and evaluate files through integrated cloud services.
- Built a Node.js provider-lookup workbench with exact identifier search, Jaro-Winkler address ranking, bulk uploads, and session-bound exports; validated authentication, isolation, input checks, and export safety with **10** local tests.
- Developed Snowflake SQL pipelines across **327K+** healthcare records, combining identifier matching, specialty joins, address comparison, and API connectivity to produce traceable extracts and quality-control reports for review.

Founder and Lead Developer | Tallyrus — **Jun 2023-Sep 2025**

- Integrated **LangChain**/OpenAI embeddings with NumPy normalization and **FAISS** similarity search, combining retrieved historical essays with teacher rubrics and prompts in GPT-4o requests to return contextual feedback for educator review.
- Implemented a Flask grading API that validates rubric, essay, prompt, and historical-example inputs before generating JSON feedback, connecting model requests to the review workflow without treating automated output as a final grade.
- Integrated a pretrained Hugging Face Transformers classifier with PyTorch, adding sentence chunking, CPU/CUDA selection, and length-weighted prediction aggregation behind a Flask API to handle longer text inputs.
- Integrated LangSmith tracing into the Tallyrus AI workflow alongside Flask grading and retrieval services, providing visibility into model interactions while keeping rubric-based feedback available for educator review.
- Connected Express/Multer upload routes for teacher files, rubrics, and historical essays to AWS S3 storage and MongoDB assignment records, making educator materials available to downstream grading and review workflows.

Data Analyst / Web Development Lead | Tech4Good at UCSC — **Mar 2023-Jun 2025**

- Implemented **20+** attributed Node/Express REST routes with MongoDB persistence and logging for chat, text-entry, and polling workflows, supporting threaded messages, anonymous responses, editing, voting, and admin controls.
- Separated chat, text-entry, and polling state across **three MongoDB databases**, using change streams, populated message logs, and persisted voter records to synchronize threads, edits, group responses, and reveal/reset operations.
- Analyzed **43,340** anonymized interaction rows across **2,805** sessions with Python/Pandas, calculating group active-time measures and building Excel/XLSX exports to turn collaboration records into structured research evidence.
- Co-authored computing-education research on undergraduate research pipelines and participatory course design, contributing to the ICER 2024 publication and the accepted, upcoming SIGCSE TS 2027 paper on collaborative curriculum design.

PROJECTS

Lume | AI Meeting Intelligence — **Dec 2025-Jan 2026**

- Designed a **RAG** engine with vector retrieval, intent routing, evidence extraction, and generator/critic refinement, combining relevant meeting and file context into citation-backed answers that users can trace to supporting sources.
- Built reusable Project Brain knowledge bases from transcripts and seed files, retaining project terminology, relationships, and contextual evidence so later questions could reuse accumulated knowledge across meetings and documents.
- Built a React and Node/Express meeting-intelligence platform with MongoDB records and AWS S3 file storage, connecting meeting capture, uploaded documents, transcript-aware search, and project questions in one reusable workflow.

Azure RAG | Document Search Proof of Concept — **Public proof of concept**

- Built a Python document-search proof of concept with Azure OpenAI embeddings, **LangChain** processing, and **FAISS** similarity search, transforming source text into retrievable vectors that support interactive questions over a document corpus.
- Implemented PDF, text, and Markdown loading with text chunking, vector indexing, and retrieval, connecting relevant source passages to citations so users could inspect the evidence supporting interactive document answers.
- Documented Azure embedding and chat-model deployment configuration for a **30+ document** corpus, providing repeatable setup for ingestion, similarity search, and cited Q&A while keeping the implementation scoped as a proof of concept.

Private Filedrop | Secure File Exchange — **Jun 2026-Aug 2026**

- Built password-protected browser uploads and API-key-protected Express APIs with OpenAPI documentation, organizing files by metadata, tags, project paths, and version history so devices and AI clients could exchange artifacts securely.
- Added local and S3-compatible storage drivers with migration tooling that preserves file bytes and metadata, retaining rollback sources and unguessable download links to support durable, versioned handoffs across computers.
- Exposed MCP tools for listing, uploading, organizing, and retrieving files, with HMAC-SHA256 webhooks and timing-safe credential checks; added deduplicated jobs, atomic claims, renewable leases, and retry limits for cross-device coordination.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7 — **Davis, CA | Graduated Jun 2025**

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu